PRACTICAL - 7 : Completely Randomized Design (CRD) - Fertilizer Analysis

PARAMETER	VALUE
n (Total observations)	25
v (Number of treatments/fertilizers)	5
Treatments	A, B, C, D, E
Replications per treatment (ri)	5

EXPERIMENTAL DATA	FERTILIZER				
	A	B	C	D	E
	8	21	17	20	17
	9	17	18	15	16
	15	12	16	13	18
	8	16	18	19	15
	10	14	19	23	13
Ti0 (Treatment total)	50	80	88	90	79
$Ti0^2/ri$	500	1280	1548.8	1620	1248.2
Treatment Mean	10	16	17.6	18	15.8

STEP-BY-STEP CALCULATIONS

STEP	CALCULATION / FORMULA	RESULT
1.	Grand Total (T00)	387
2.	Correction Factor $CF = T00^2/n$	5990.76
3.	Sum of $Ti0^2/ri$	6197
4.	$SST = Sum(Ti0^2/ri) - CF$	206.24
5.	Sum of all Yij^2	6361
6.	$TSS = Sum(Yij^2) - CF$	370.24
7.	SSE = TSS - SST	164
8.	df (treatment) = v - 1	4
9.	df (error) = n - v	20
10.	MST = SST / df(treatment)	51.56
11.	MSE = SSE / df(error)	8.2
12.	F-value = MST / MSE	6.287804878
13.	F-critical (4,20) at α=0.05	2.866

ANOVA TABLE

Source of Variation	df	SS	MS	F-value
Treatment	4	206.24	51.56	6.287804878
Error	20	164	8.2	
Total	24	370.24		

PAIRWISE COMPARISON (t-test)

t-critical (20) at α=0.025 (two-tailed) 2.086

Comparison	Mean Diff	t-statistic	Result
A vs B	-6	-3.312945782	Reject
A vs C	-7.6	-4.196397991	Reject
A vs D	-8	-4.417261043	Reject
A vs E	-5.8	-3.202514256	Reject
B vs C	-1.6	-0.8834522086	Accept
B vs D	-2	-1.104315261	Accept
B vs E	0.2	0.1104315261	Accept
C vs D	-0.4	-0.2208630521	Accept
C vs E	1.8	0.9938837347	Accept
D vs E	2.2	1.214746787	Accept

RESULTS

RESULT	VALUE
F-calculated	6.287804878
F-critical (4,20)	2.866
Decision	Reject H0 - Treatments differ significantly
Conclusion	Fertilizer A shows significantly lower yields compared to others

PRACTICAL - 8 : Completely Randomized Design (CRD) - Nitrogen Losses Analysis

GIVEN VALUES				
PARAMETER	**VALUE**			
n (Total observations)	30			
v (Number of treatments)	4			
Treatments	Continuous small grain, Alternate small shell, Small grain manure, Small grain peas			
Study period	20 years in Kansas			

EXPERIMENTAL DATA - Nitrogen Losses				
Obs	**Continuous small grain**	**Alternate small shell**	**Small grain manure**	**Small grain peas**
1	0.012	0.011	0.019	0.006
2	0.019	0.018	0.026	0.02
3	0.028	0.033	0.008	0.028
4	0.017	0.03	0.032	0.012
5	0.017	0.029	0.011	0.009
6		0.022	0.018	0.01
7		0.022	0.018	0.01
8		0.029	0.031	0.015
9			0.018	
ri (Replications)	5	8	9	8
Ti0 (Treatment total)	0.093	0.194	0.181	0.11
$Ti0^2$	0.008649	0.037636	0.032761	0.0121
$Ti0^2/ri$	0.0017298	0.0047045	0.003640111111	0.0015125

STEP-BY-STEP CALCULATIONS		
STEP	**CALCULATION / FORMULA**	**RESULT**
1.	Grand Total (T00)	0.578
2.	Correction Factor $CF = T00^2/n$	0.01113613333
3.	Sum of $Ti0^2/ri$	0.01158691111
4.	$SST = Sum(Ti0^2/ri) - CF$	0.0004507777778
5.	Sum of all Yij^2	0.013
6.	$TSS = Sum(Yij^2) - CF$	0.001863866667
7.	SSE = TSS - SST	0.001413088889
8.	df (treatment) = v - 1	3
9.	df (error) = n - v	26
10.	MST = SST / df(treatment)	0.0001502592593
11.	MSE = SSE / df(error)	0.00005434957265
12.	F-value = MST / MSE	2.764681522
13.	F-critical (3,26) at α=0.05	2.975
14.	p-value	0.06204066755

ANOVA TABLE				
Source of Variation	**df**	**SS**	**MS**	**F-value**
Treatment	3	0.0004507777778	0.0001502592593	2.764681522
Error	26	0.001413088889	0.00005434957265	
Total	29	0.001863866667		

RESULTS				
RESULT	**VALUE**			
F-calculated	2.764681522			
F-critical (3,26)	2.975			
p-value	0.06204066755			
Decision	Accept H0 - No significant difference			
Conclusion	Different crop rotations do not have significant effect on nitrogen losses (p > 0.05)			

PRACTICAL - 9 : Randomized Block Design (RBD) - Variety Analysis

GIVEN VALUES

PARAMETER	VALUE
v (Number of varieties/treatments)	8
b (Number of blocks)	5
n (Total observations)	40
Design	Randomized Block Design

EXPERIMENTAL DATA

Variety	Block 1	Block 2	Block 3	Block 4	Block 5	Total (Ti0)
1	296	357	340	331	348	1672
2	402	390	431	340	320	1883
3	437	334	426	320	296	1813
4	303	319	310	260	242	1434
5	469	405	442	487	394	2197
6	345	342	358	300	308	1653
7	324	339	357	352	220	1592
8	488	374	401	338	320	1921
Block Total (Toj)	3064	2860	3065	2728	2448	14165

STEP-BY-STEP CALCULATIONS

STEP	CALCULATION / FORMULA	RESULT
1.	Grand Total (T00)	14165
2.	Correction Factor CF = T00²/(v*b)	5016180.625
3.	Sum of Ti0² / b (varieties)	5093704.2
4.	SST (Treatment) = Sum(Ti0²/b) - CF	77523.575
5.	Sum of Toj² / v (blocks)	5049576.125
6.	SSB (Blocks) = Sum(Toj²/v) - CF	33395.5
7.	Sum of all Yij²	5164533
8.	TSS = Sum(Yij²) - CF	148352.375
9.	SSE = TSS - SST - SSB	37433.3
10.	df (treatment) = v - 1	7
11.	df (blocks) = b - 1	4
12.	df (error) = (v-1)(b-1)	28
13.	MST = SST / df(treatment)	11074.79643
14.	MSB = SSB / df(blocks)	8348.875
15.	MSE = SSE / df(error)	1336.903571
16.	F-value (treatment) = MST / MSE	8.283915658
17.	F-value (blocks) = MSB / MSE	6.244934323
18.	F-critical (7,28) at α=0.05	2.359
19.	F-critical (4,28) at α=0.05	2.714

ANOVA TABLE

Source of Variation	df	SS	MS	F-value	F-critical
Treatment (Varieties)	7	77523.575	11074.79643	8.283915658	2.359
Blocks	4	33395.5	8348.875	6.244934323	2.714
Error	28	37433.3	1336.903571		
Total	39	148352.375			

EFFICIENCY COMPARISON: RBD vs CRD

20.	For CRD: Calculate overall variance S²	3803.907051
21.	V(y_mean) for CRD = (1/n - 1/N)*S²	380.3907051
22.	V(y_mean) for RBD = MSE/b	267.3807143
23.	Efficiency = V(CRD) / V(RBD)	1.422655729
24.	Alternative: Efficiency = (MSB*(b-1) + MSE*(v-1)(b-1)) / (MSE*(v*b-1))	1.358454802

PAIRWISE COMPARISON (t-test)

t-critical (28) at α=0.025 (two-tailed)		2.048
Standard Error for comparison		23.12490927

Comparison	Mean Diff	t-statistic	Result
Variety 1 vs 2	-42.2	-1.824872025	Accept
Variety 1 vs 5	-105	-4.540558356	Reject
Variety 2 vs 3	14	0.6054077808	Accept
Variety 4 vs 5	-152.6	-6.598944811	Reject
Variety 5 vs 6	108.8	4.704883325	Reject
Variety 5 vs 8	55.2	2.387036393	Reject

RESULTS

RESULT	VALUE		
F-calculated (Treatment)	8.283915658		
F-critical (7,28)	2.359		
F-calculated (Blocks)	6.244934323		
F-critical (4,28)	2.714		
Decision (Treatment)	Reject H0 - Varieties differ significantly		
Decision (Blocks)	Reject H0 - Blocks differ significantly		
Efficiency (RBD vs CRD)	1.422655729		
Conclusion	RBD is more efficient than CRD. Varieties show significant differences.		